Course: Foundations of AI with Embeddings

# Lecture 1: Basic dimensionality reduction algorithms

PCA · MDS · Local MDS · Isomap

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Constructor University, 2026

## Outline

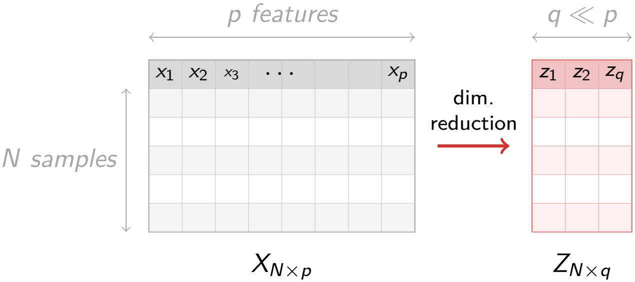
# 1. Problem Statement

## Problem Statement

- We have data in a $p$-dimensional space — too complex to visualise or analyse directly.
- We want to transform points from this high-dimensional space into a lower-dimensional subspace (e.g. 2D).
- Why?
  - After reduction we can find and visualise clusters, detect outliers.
  - Reducing noise in the data.
  - Data compression.

**Key question:** How to choose the optimal projection?

# From $N \times p$ to $N \times q$




*$X$ — original data matrix; $Z$ — low-dimensional representation; each **row** is one sample*

# 2. Principal Component Analysis (PCA)

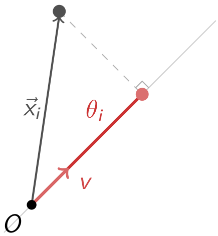
# PCA — Idea

- **Informativeness $\approx$ variance.** A feature with constant values carries no information and can be discarded.
- We want to project data onto a subspace where the variance along each basis direction is **maximised**.
- The dimensionality $q$ of the target subspace is a hyperparameter chosen by the user.

## Variance along an axis

Let $v \in \mathbb{R}^p$ be a unit vector we project onto. Then

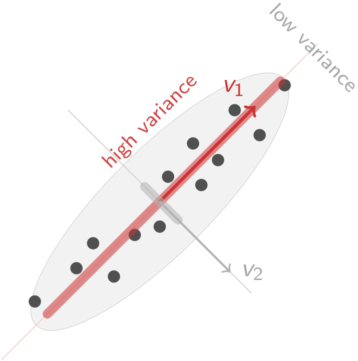
- $\theta_i := \|z_i\| = \frac{v^T \cdot x_i}{\|v\|} = v^T \cdot x_i$
- $Var(\theta) := \frac{1}{N}\sum_{i=1}^{N}(\theta_i - \bar{\theta})^2 = \frac{1}{N}\sum_{i=1}^{N}(v^T \cdot x_i - \bar{\theta})^2$
- Assume that $\bar{\theta} = 0$ (we can always subtract the mean from the data).
- $Var(\theta) = \frac{1}{N}\sum_{i=1}^{N}(v \cdot x_i)^2 = \frac{1}{N}(Xv)^T(Xv) = v^T \Sigma v$,
  where $\Sigma = \frac{1}{N}X^T X$ is the covariance matrix.




Therefore:

$$\hat{v} = \arg\max_v v^T \Sigma v$$

# PCA: Solution




**Solution:** leading $q$ eigenvectors of $\Sigma$:

$$\Sigma\,\hat{v}_1 = \lambda_1\,\hat{v}_1,$$

$$\cdots$$

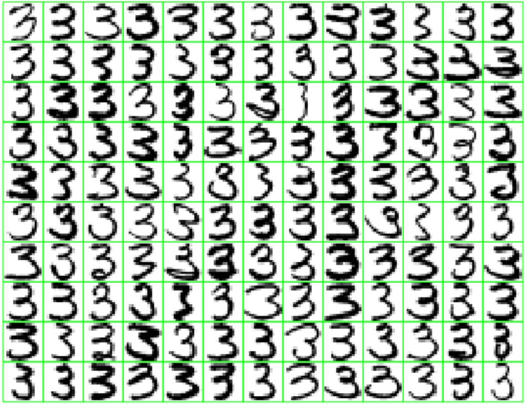
$$\Sigma\,\hat{v}_q = \lambda_q\,\hat{v}_q,$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_q$.

**For $q$ axes:** top-$q$ eigenvectors (mutually orthogonal).

# A case study: handwritten digits

- 130 hadwritten threes.
- Each image is of $16 \times 16 = 256$ pixels. Feature is a pixel's grayscale value (0..255).



# Shortcomings of global approaches

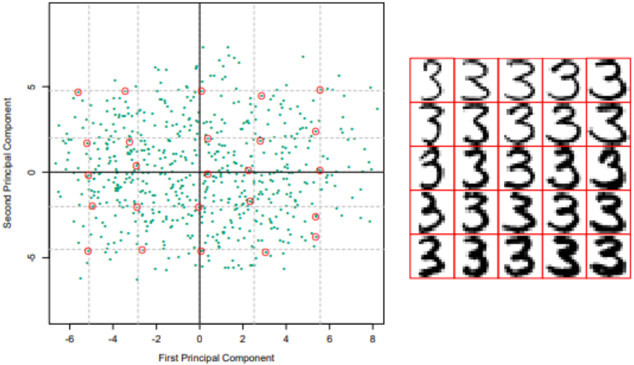



**FIGURE 14.23.** *(Left panel:) the first two principal components of the hand-written threes. The circled points are the closest projected images to the vertices of a grid, defined by the marginal quantiles of the principal components. (Right panel:) The images corresponding to the circled points. These show the nature of the first two principal components.*

# 3. Multidimensional Scaling (MDS)

# MDS — Idea

- Often we only care about preserving **pairwise distances**, not the absolute coordinates.
- MDS finds $N$ points $z_1, \ldots, z_N \in \mathbb{R}^q$ minimising the **stress function**:

$$S_M(z_1, \ldots, z_N) = \sum_{i<i'} \left(d_{ii'} - \|z_i - z_{i'}\|\right)^2 \tag{3}$$

  where $d_{ii'}$ are the original pairwise distances.
- MDS is a **non-linear** method (despite a linear-algebra solution for classical MDS).

# MDS — Similarity-Based Variant

In practice, distances can be replaced by **similarities**:

$$s_{ii'} = \langle x_i - \bar{x},\ x_{i'} - \bar{x} \rangle,$$

and the stress becomes:

$$S_C(z_1, \ldots, z_N) = \sum_{i<i'} \big( s_{ii'} - \langle z_i - \bar{z},\ z_{i'} - \bar{z} \rangle \big)^2.$$

## Strengths

- Non-linear method
- Preserves **global** pairwise (Euclidean) distances

## Limitations

- Still uses global Euclidean distances
- Does not capture **local** structure of the manifold
- Sensitive to outliers in the distance matrix

# 4. Local MDS and ISOMAP

## Motivation: Global vs. Local Structure

- PCA and MDS preserve **global** Euclidean distances.
- For data lying on a **curved manifold**, Euclidean distances are misleading: two points on opposite sides of a roll are *far* along the surface but *close* in Euclidean space.
- We want methods that respect the **intrinsic geometry** of the data.

**Idea**

Replace Euclidean distances with distances that follow the shape of the manifold.

# Shortcomings of global approaches

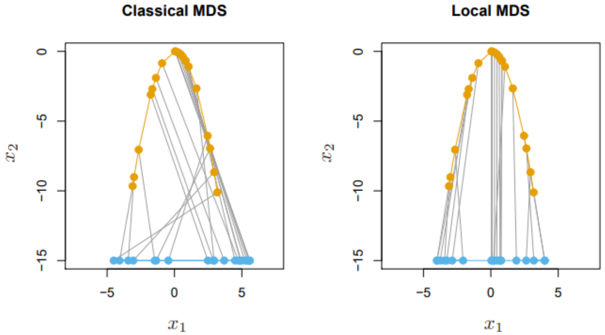



**FIGURE 14.44.** *The orange points show data lying on a parabola, while the blue points shows multidimensional scaling representations in one dimension. Classical multidimensional scaling (left panel) does not preserve the ordering of the points along the curve, because it judges points on opposite ends of the curve to be close together. In contrast, local multidimensional scaling (right panel) does a good job of preserving the ordering of the points along the curve.*

# Local MDS

Let $\mathcal{N}$ be the set of *neighbour pairs*: $(i, i') \in \mathcal{N}$ iff $i'$ is among the $K$ nearest neighbours of $i$.

Modified stress function:

$$S_L = \underbrace{\sum_{(i,i') \in \mathcal{N}} (d_{ii'} - \|z_i - z_{i'}\|)^2}_{\text{local: preserve neighbour distances}} + w \underbrace{\sum_{(i,i') \notin \mathcal{N}} (D - \|z_i - z_{i'}\|)^2}_{\text{repulsion: push non-neighbours apart}}$$

where $D$ is a large constant and $w$ is a weight.

**Practical relaxation** (replacing the repulsion term):

$$S_L \approx \sum_{(i,i') \in \mathcal{N}} (d_{ii'} - \|z_i - z_{i'}\|)^2 + \tau \sum_{(i,i') \notin \mathcal{N}} \|z_i - z_{i'}\|,$$

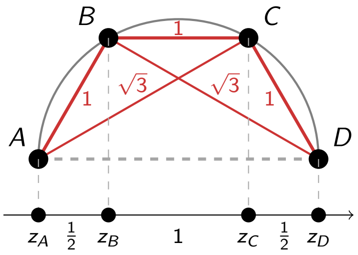
where $\tau = 2wD$ is a tunable regularisation parameter.

## Local MDS: Worked Example — Stress calculation ($K=2,\ w=1,\ D=3$)

**Points on unit semicircle:**

$A=(-1,0),\ \ B=\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right)$

$C=\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right),\ \ D=(1,0)$




| Pair | $d_{ij}$ | $\lvert z_i - z_j \rvert$ | $\in \mathcal{N}$ |
|---|---|---|---|
| $AB$ | 1 | $1/2$ | ✓ |
| $BC$ | 1 | 1 | ✓ |
| $CD$ | 1 | $1/2$ | ✓ |
| $AC$ | $\sqrt{3}$ | $3/2$ | ✓ |
| $BD$ | $\sqrt{3}$ | $3/2$ | ✓ |
| $AD$ | 2 | 2 | × |

$$S_L = \frac{1}{4} + 0 + \frac{1}{4} + 2\cdot\left(\sqrt{3}-\frac{3}{2}\right)^2 + 1\cdot(3-2)^2$$

$$\boxed{S_L \approx 1.608}$$

# ISOMAP

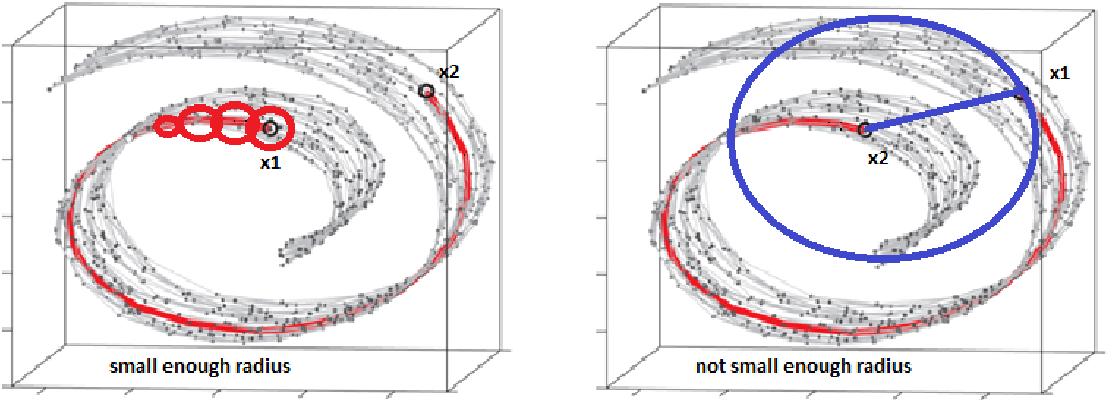
1. **Build a neighbourhood graph.** Connect each point to its $K$ nearest neighbours (or all points within radius $\varepsilon$). Edge weights $=$ Euclidean distances.

2. **Compute geodesic distances.** Shortest-path distances on the graph (e.g. via Dijkstra's algorithm) approximate *manifold* distances $d_{ii'}$.

3. **Apply MDS.** Use the geodesic distance matrix as input to standard MDS.

# ISOMAP

RADIUS IS FIXED



# Beyond Isomap: Modern Manifold Methods

**t-SNE** van der Maaten & Hinton, 2008

- Accounts for **local density**, not just distances
- Very effective for 2D/3D visualisation
- **Slow:** $\mathcal{O}(N^2)$ — scales poorly

**UMAP** McInnes et al., 2018

- Preserves both **local and global** structure
- Stronger theoretical foundations (topology)
- **Faster** and more scalable than t-SNE

These methods will be covered in detail in the next lectures.

## References


- Golub, G.H., Van Loan, C.F. *Matrix Computations*. Johns Hopkins University Press, 4th ed., 2013.
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